

HW2.1

SCIE 001 Math

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1. Suppose that f is differentiable with $f' > 0$ for all x and suppose that $f(1) = 0$. Set

$$F(x) = \int_0^x f(t) \, dt$$

Determine if each statement is True or False. Justify your answer.

- (a) The function $F(x)$ is continuous.
- (b) The function $F(x)$ is not twice differentiable.
- (c) The value $x = 1$ is a critical point of F .
- (d) The function $F(x)$ takes on a local minimum at $x = 1$.
- (e) $F(1) > 0$.

Based on your answers above, make a rough sketch of the graph of $F(x)$.

2. Recall in class we proved that $\sum_{i=1}^k i = \frac{k(k+1)}{2}$. In this problem, we will construct a proof of

$$S = \sum_{i=1}^k i^2 = \frac{k(k+1)(2k+1)}{6}$$

- (a) Calculate the value of S for $k = 1, 2, 3$ and 4.
- (b) Calculate the value of $\frac{k(k+1)(2k+1)}{6}$.
- (c) Show that $\sum_{i=1}^k i^2 = \frac{k(k+1)(2k+1)}{6}$ by constructing a proof by induction by:
 - i. showing the equality is true for the base case (i.e., $k = 1$).
 - ii. for each positive integer N , the equality holds for $k = N$ implies the equality for $k = N + 1$. Be sure to end your proof with an appropriate conclusion statement (i.e., "Therefore, by induction, ...").

3. where P_{in} and P_{out} denote the power corresponding to, respectively, energy flowing in from the Sun and energy flowing out as the Earth's thermal radiation.

4. **Your task:** Derive the EBM given above and verify that the units on both sides of the

Left Side	Right Side
$= JK^{-1} \frac{K}{s}$	$= W - W$
$= \frac{J}{s}$	$= W$
$= W$	

$$W = W$$

Left Side = Right Side

Therefore, the units are consistent.

$$\lim_{T \rightarrow 247^-} \alpha(T) = \alpha(247) \quad \lim_{T \rightarrow 247^+} \alpha(T) = \alpha(247) \quad \lim_{T \rightarrow 282^-} \alpha(T) = \alpha(282) \quad \lim_{T \rightarrow 282^+} \alpha(T) = \alpha(282)$$

5. **Your task:** Whi

A Guide to Finding Intersection Points of P_{in} and P_{out} in Python

Create a new Jupyter notebook using anaconda. Copy and paste the following lines of code into a new cell.